Compact–weakly mixing decomposition

This is intended to be a short, self-contained introduction to the compact–weakly mixing decomposition in unitary actions of countable amenable groups (amenability is only needed due to the definition we give of weakly mixing vectors). All the material in these notes is well-known. For further details, see for example:

- D. Kerr and H. Li, *Ergodic Theory: Independence and Dichotomies*, Springer, 2016.
- K. Petersen, *Ergodic Theory*, Cambridge University Press, 1983.

Let $(U_g)_{g\in G}$ be a unitary representation of a countable amenable group G on a real or complex Hilbert space H. We will think of H as a metric space and U_g as linear isometries.

Compact vectors

Definition 1. We say $v \in H$ is compact if the closure of its orbit $\overline{Gv} := \overline{\{gv; g \in G\}}$ is compact in the norm topology. We say the representation U is compact if all $v \in H$ are compact.

If H is finite-dimensional, then all $v \in H$ are compact, as their orbits are contained in bounded balls. For example, a rotation of $\mathbb{S}^1 \subseteq \mathbb{C}$ or any group of isometries of the spheres $\mathbb{S}^n$, $n \geq 2$ (e.g. the group of isometries of higher-dimensional simplices) can be seen as compact actions.

Proposition 2. *Compact vectors form a closed, U-invariant subspace of H.*

Proof sketch. If v, w are compact, then so are scalar multiples λv (as $G(\lambda v) = \lambda(Gv)$) and $v+w$ (as $G(v+w) \subseteq Gv+Gw$). If $v_n \to v$ and v_n are compact, then one can readily check that Gv is totally bounded, so its closure is compact. U-invariance follows from the definition of compact vector. □

Theorem 3. *If $v \in H$ is compact and $\varepsilon > 0$, then the set of returns*

$$R = \{g \in G; \|U_g v - v\| < \varepsilon\} \tag{1}$$

is two-sided syndetic, i.e. there exists a finite set F such that $RF = FR = G$.

Even better, the set R in Equation (1) is a *Bohr neighborhood of the identity* (if $G = \mathbb{Z}$, R contains something of the form $\{n \in \mathbb{Z}; \|nv\|_{\mathbb{T}^d} < \delta\}$ for some $d \in \mathbb{N}$, $v \in \mathbb{T}^d$ and $\delta > 0$).

Proof. As the orbit Gv is precompact, there is a finite subset F of G such that Fv is ε-dense in Gv, that is, $d(w, Fv) < \varepsilon$ for all $w \in Gv$. Therefore, as for all $g \in G$ U_g is an isometry of Gv, $U_g(Fv) = (gF)v$ is also ε-dense in Gv for all $g \in G$. In particular, gF intersects R for all $g \in G$, so $RF^{-1} = G$. Also note that $R = R^{-1}$, because $d(v, U_g v) < \varepsilon$ iff $d(v, U_g^{-1} v) < \varepsilon$, so $FR = (RF^{-1})^{-1} = G$. We can take the finite set in the theorem to be $F \cup F^{-1}$. □

More generally, the visits of v to any nonempty open subset of Gv are two-sided syndetic.

Corollary 4. *If $(X, \mathcal{B}, \mu, T)$ is a m.p.s.*[1]*, $A \in \mathcal{B}$ and 1_A is a compact vector, then for every $\varepsilon > 0$ the set $\{n \in \mathbb{Z}; \mu(A \cap T^n A) > \mu(A) - \varepsilon\}$ is two-sided syndetic (and in fact a Bohr nhood of 0).*

Proof. Any vector v close enough to 1_A will satisfy $\mathrm{Re}\langle 1_A, v\rangle > \mu(A) - \varepsilon$. □

[1] Here, by m.p.s. we mean **invertible** probability measure-preserving system.

Weakly mixing vectors

Definition 5. We say $w \in H$ is weakly mixing if for all $v \in H$, UC-lim $|\langle v, U_g w\rangle| = 0$. That is, for any left or right Følner sequence (F_N) in G,

$$\lim_N \frac{1}{|F_N|} \sum_{g \in F_N} |\langle v, U_g w\rangle| = 0. \tag{2}$$

We say $(U_g)_{g\in G}$ is weakly mixing if all $v \in H$ are weakly mixing.

Equivalently, w is weakly mixing if, for all $v \in H$ and $\varepsilon > 0$, the set

$$\{g \in G; |\langle v, U_g w\rangle| > \varepsilon\}$$

has upper Banach density 0 (i.e. it has density 0 in all Følner sequences).

That is, for each fixed $v \in H$, the vector $U_g w$ is almost orthogonal to v for most $g \in G$.

Proposition 6. *Weakly mixing vectors form a closed subspace of H.*

Proof. It is clear that if w_1, w_2 are weakly mixing, then so are λw_1 and $w_1 + w_2$, using that for all $v \in H$ and $g \in G$,

$$|\langle v, U_g(w_1 + w_2)\rangle| \le |\langle v, U_g(w_1)\rangle| + |\langle v, U_g(w_2)\rangle|.$$

Moreover, if $w_n \to w$ and w_n are weakly mixing, then for any fixed $v \in H$, the fact that Equation (2) holds for all w_n implies that it holds for w. □

Note that if $v \neq 0$ is weakly mixing, then we can construct by recursion a sequence $g_1, g_2, g_3, \dots$ such that $\lim_{m,n\to\infty}\langle U_{g_n} v, U_{g_m} v\rangle = 0$: given $g_1, \dots, g_{n-1}$, pick g_n outside of the set

$$\left\{ g \in G; |\langle U_g v, U_{g_i} v\rangle| \ge \frac{1}{n} \text{ for some } i = 1, \dots, n-1 \right\},$$

which density 0 in all Følner sequences. The family $\{g_1 v, g_2 v, \dots\}$ is clearly not precompact.

As $\langle v, U_g w\rangle = \langle U_{g^{-1}} v, w\rangle$ for all $g \in G$ and $v, w \in H$, a vector w is weakly mixing iff for all $v \in H$ we have

$$\lim_N \frac{1}{|F_N|} \sum_{g \in F_N} |\langle U_g v, w\rangle| = 0. \tag{3}$$

That is, w is 'almost orthogonal to any given orbit'. This immediately implies the following:

Proposition 7. *If $v \in H$ is compact and $w \in H$ is weakly mixing, then $v \perp w$.*

Proof. If $\lambda := |\langle v, w\rangle| > 0$, then by compactness of v the set $\left\{g \in G; |\langle U_g v, w\rangle| > \frac{\lambda}{2}\right\}$ is syndetic. But that directly contradicts Equation (3). □

Measurable viewpoint. For invertible measure preserving transformations $(X, \mathcal{B}, \mu, T)$, working with complex-valued functions, we have

$$L^2(X, \mu) = \mathbb{C}1_X \oplus L^2_0(X, \mu),$$

where $L^2_0(X, \mu)$ are the functions with integral 0. We say T is weakly mixing when the action of U_T on $L^2_0(X, \mu)$ is weakly mixing.

Compact–weakly mixing decomposition

Theorem 8. *Let H_c, H_{wm} denote the closed subspaces of compact and weakly mixing vectors, respectively. Then*

$$H = H_{\mathrm{c}} \oplus H_{\mathrm{wm}},$$

both H_{c} and H_{wm} are U-invariant, and H_{c} is an orthogonal direct sum of finite-dimensional U-invariant subspaces.

Corollary 9 (Dichotomy of positive returns)**.** *For any $v \in H$, one of the following two things happens:*

1. *For all $\varepsilon > 0$, the set $\{g \in G; \mathrm{Re}\langle v, U_g v\rangle > \varepsilon\}$ has density 0 in any left or right Følner sequence.*
2. *For small enough $\varepsilon > 0$, the set $\{g \in G; \mathrm{Re}\langle v, U_g v\rangle > \varepsilon\}$ is two-sided syndetic, i.e. it has positive upper density in all left and right Følner sequences.*

Proof. Write $v = v_c + v_{\mathrm{wm}}$. By orthogonality and invariance,

$$\langle v, U_g v\rangle = \langle v_c, U_g v_c\rangle + \langle v_{\mathrm{wm}}, U_g v_{\mathrm{wm}}\rangle.$$

If $v_c = 0$, the first alternative follows from weak mixing. Otherwise, put $a = \|v_c\|^2 > 0$. Compactness implies that

$$R = \{g \in G; \mathrm{Re}\langle v_c, U_g v_c\rangle > 3a/4\}$$

is two-sided syndetic, while weak mixing implies that

$$B = \{g \in G; |\langle v_{\mathrm{wm}}, U_g v_{\mathrm{wm}}\rangle| \geq a/4\}$$

has density zero along every left and right Følner sequence. Consequently, $R \setminus B$ has positive lower density along every such sequence, since R does. Hence $R \setminus B$ is two-sided syndetic: failure of either syndeticity condition would allow translates of Følner sets inside its complement, giving a Følner sequence along which its density is zero. Finally, $\mathrm{Re}\langle v, U_g v\rangle > a/2$ on $R \setminus B$, proving the second alternative for every $0 < \varepsilon \leq a/2$. □

We will deduce Theorem 9 from the following proposition:

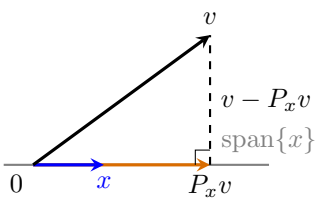
Proposition 10. *If some vector $w \in H$ is not weakly mixing, then H has a nonzero finite-dimensional U-invariant subspace.*

Indeed, choose a maximal orthogonal family of nonzero finite-dimensional U-invariant subspaces, and let K be their closed span. Applying the claim to the restricted representation on $K^\perp$ shows that every vector in $K^\perp$ is weakly mixing, since otherwise we could enlarge the family. As $K \subseteq H_c$ and $H_c \perp H_{\mathrm{wm}}$, it follows that $K = H_c$ and $K^\perp = H_{\mathrm{wm}}$.

Proof of Proposition 11.
We will use that if $v, x \in H$ with $\|x\| = 1$, then

$$|\langle v, x\rangle|^2 = \langle v, P_x v\rangle,$$

where P_x is the orthogonal projection onto the subspace (line) generated by x.



We may assume H is separable, otherwise take the smallest closed invariant subspace containing w. Suppose $\|w\| = 1$. As w is not w.m., there are $v \in H$ and a left Følner sequence (F_N) (or right Følner, in which case we can take its inverse) such that

$$0 < \lim_N \frac{1}{|F_N|} \sum_{g \in F_N} |\langle v, U_g w\rangle|^2 = \lim_N \frac{1}{|F_N|} \sum_{g \in F_N} \langle v, P_{gw} v\rangle = \lim_N \left\langle v, \left(\frac{1}{|F_N|} \sum_{g \in F_N} P_{gw} \right) v \right\rangle \quad (4)$$

Now, as the operators $A_N := \frac{1}{|F_N|} \sum_{g \in F_N} P_{gw}$, $N \in \mathbb{N}$, have norm ≤ 1, after taking a subsequence they must converge in the weak operator topology to some operator A. This operator A is compact (in fact, it has Hilbert-Schmidt norm ≤ 1, because the A_N do). And by Equation (4), $\langle v, Av\rangle > 0$, so $A \neq 0$. This implies that A has some nonzero eigenvalue, λ (in fact, $\lambda > 0$; this is because A is positive, as the limit of the positive operators A_N).

Let H_λ be the λ-eigenspace of A; as A is compact, H_λ must be finite-dimensional, and we claim it is invariant under G. It is enough to prove that A commutes with U_h for all $h \in G$, as then $Ax = \lambda x \iff A(U_h x) = \lambda U_h x$ for all x.

And indeed, given $g \in G$, as $P_{gw} = U_g P_w U_g^{-1}$ (both operators are projections to a line, and both fix gw), we have $A_N = \frac{1}{|F_N|} \sum_{g \in F_N} U_g P_w U_g^{-1}$, so for all $h \in G$ we have

$$\begin{aligned}
\|U_h A U_h^{-1} - A\| &\leq \liminf_{N \to \infty} \|U_h A_N U_h^{-1} - A_N\| \\
&= \liminf_{N \to \infty} \left\| \frac{1}{|F_N|} \sum_{g \in F_N} U_h U_g P_w U_g^{-1} U_{h^{-1}} - \frac{1}{|F_N|} \sum_{g \in F_N} U_g P_w U_g^{-1} \right\| \\
&= \liminf_{N \to \infty} \left\| \frac{1}{|F_N|} \sum_{g \in hF_N} U_g P_w U_g^{-1} - \frac{1}{|F_N|} \sum_{g \in F_N} U_g P_w U_g^{-1} \right\| \\
&\leq \liminf_{N \to \infty} \frac{1}{|F_N|} \sum_{g \in hF_N \Delta F_N} \|U_g P_w U_g^{-1}\| = \liminf_{N \to \infty} \frac{|hF_N \Delta F_N|}{|F_N|} = 0. \qquad \square
\end{aligned}$$

Case $G = \mathbb{Z}$

Over a complex Hilbert space, every finite-dimensional unitary representation of an abelian group G admits an orthonormal basis of common eigenvectors, by the simultaneous diagonalization theorem for commuting normal operators. Hence every finite-dimensional irreducible representation is one-dimensional; for $G = \mathbb{Z}$, these are precisely the lines spanned by eigenvectors of U_1. (Over real Hilbert spaces, irreducible representations can also be two-dimensional rotations.)

Theorem 11 (Compact-weakly mixing decomposition in $\mathbb{Z}$)**.** *For any unitary action of $\mathbb{Z}$ on a complex Hilbert space, we have*

$$H = H_{\mathrm{c}} \oplus H_{\mathrm{wm}},$$

both H_{c} and H_{wm} are U-invariant, and H_{c} is an orthogonal direct sum of eigenspaces.

Continuous spectrum and discrete spectrum

We keep $G = \mathbb{Z}$, write $U = U_1$, and work over a complex Hilbert space, with the inner product linear in the first argument. (For a real Hilbert space, pass to its complexification.) Recall Herglotz' theorem: for every $v \in H$, there is a unique finite positive Borel measure σ_v on $\mathbb{S}^1$ such that

$$\langle U^n v, v\rangle = \int_{\mathbb{S}^1} z^n \, d\sigma_v(z), \qquad n \in \mathbb{Z}.$$

σ_v is called the *spectral measure* of v; its total mass is $\|v\|^2$.

The compact–weakly mixing decomposition has the following spectral interpretation:

$$v \in H_c \iff \sigma_v \text{ is discrete (or } \textit{purely atomic}),$$
$$v \in H_{\mathrm{wm}} \iff \sigma_v \text{ is continuous (or } \textit{atomless}).$$

Therefore, if $(X, \mathcal{B}, \mu, T)$ is a m.p.s. with Koopman operator $U_T f = f \circ T$ on $L^2(X, \mu; \mathbb{C})$, then

$$T \text{ has discrete spectrum} \iff U_T \text{ is compact} \iff \sigma_f \text{ is discrete for every } f \in L^2.$$

$$T \text{ is weakly mixing} \iff U_T|_{L^2_0} \text{ is weakly mixing} \iff \sigma_f \text{ is continuous for every } f \in L^2_0.$$

More precisely, if $v = v_c + v_{\mathrm{wm}}$ is the compact–weakly mixing decomposition of v, then

$$\sigma_v = \sigma_{v_c} + \sigma_{v_{\mathrm{wm}}},$$

where σ_{v_c} and $\sigma_{v_{\mathrm{wm}}}$ are exactly the discrete and continuous parts of σ_v, respectively.